Final Project

1 Part 1 : Programming

1.1 Snapshot of columns & cleaned types

Table 1: First six rows of the dataset

ID	Gender	Height	Weight	Physical.Activity
ID1	Male	172.83	62.79	Intense
ID2	Female	165.72	74.85	Intense
ID3	Female	169.44	54.56	Moderate
ID4	Female	169.61	66.93	Intense
ID5	Male	183.50	55.31	None
ID6	Male	174.97	76.28	Moderate

1.2 Abstract

Using a sample of 1,000 adults, we examined three questions:

i. Whether weight is linearly related to height;
ii. Whether mean height differs between males and females; and
iii. Whether gender and physical activity are associated.

For Q1, a simple linear regression of Weight on Height estimates that each additional centimetre of height corresponds to about **0.70 kg** higher expected weight ($\mathbf{R^2 = 20.6\%}$, p $\mathbf{7.6\times10^{\ 2}}$). This model captures a clear trend but leaves substantial unexplained variability, indicating other factors (e.g., age, body composition) matter.

For Q2, an equal-variance two-sample t-test shows males are taller on average by **10.26 cm** (95% CI **9.77–10.76 cm**, p $\mathbf{7.98\times10^{\ 21}}$). The test targets a difference in means, not distributional shape; conclusions rely on approximate normality and similar variances, which diagnostics supported.

For Q3, a chi-square test on the **Gender** $\times$ **Physical Activity** contingency table detects **no evidence of association** (p **0.306**); expected counts comfortably exceed 5, so the test is appropriate, but it is insensitive to ordered intensity or subtle effect sizes. Overall, height explains a modest share of weight variation, males are substantially taller than females, and activity levels appear similar across genders under this design.

1.3 Introduction

"I am aiming for a High Distinction for this part"

This report analyses a class dataset of **1,000 adults** containing five variables: an identifier (**ID**), **Gender** (Female/Male), **Height** (cm), **Weight** (kg), and **Physical Activity** level (None/Moderate/Intense). We address three practical questions:

1. **Height–Weight relation:** Is body weight linearly related to height?
2. **Gender difference in mean height:** Do males and females differ in average height?
3. **Association between Gender and Activity:** Are activity levels distributed differently across genders?
4. The analyses are intentionally simple and transparent. For Q1 we fit a **simple linear regression** of Weight on Height to estimate the average change in weight per additional cm of height and to quantify explained variation. For Q2 we use a **two-sample *t*-test with equal variances** to compare mean heights between genders. For Q3 we apply a **chi-square test of independence** to the Gender $\times$ Activity contingency table. All decisions use **= 0.05** and are supported by routine diagnostics (residual checks, QQ plots, expected counts).

These choices match the data structure (one row per person, mostly continuous or categorical variables) and the project brief. While the models are appropriate for the questions posed, they do not control for potential confounders (e.g., age or body composition), so results speak to **associations** rather than causation. The report is fully reproducible: all code runs from a parameterised Quarto file, with figures generated directly from the data.
This report uses the class dataset [@project2025data].

1.4 Methodology

- **Q1:** Simple linear regression `Weight ~ Height`. Assumptions: linearity, independent errors, constant variance, normal residuals. (R Core Team 2025)
- **Q2:** Two-sample t-test (equal variances) for mean **Height** across **Gender**.
- **Q3:** Chi-square test of independence for **Gender** $\times$ **Physical Activity**.
- Decision rule: reject H if p-value < 0.05.

1.4.0.1 Implementation notes (Part II)

To keep the analysis reproducible and concise, we implemented a small set of helper functions in a separate script (`analysis_wrappers.R`):

- `lm_height_weight(df)` — fits Weight ~ Height and returns slope, 95% CI, R^2, p.
- `compare_mean_height(df)` — two-sample t-test Male vs Female with plot.
- `assoc_gender_activity(df)` — ^{2} test of Gender × Physical Activity with observed/expected tables.

We also defined a simple S3 `print()` method to emit a one-line, human-readable summary for each result object. All figures and tables in Results are produced by calling these functions on the cleaned `data_raw`.

1.5 Results

1.5.1 Analysis 1 — Is weight linearly related to height?

Hypotheses (= 0.05)

H0: The slope of weight on height is 0 (no linear relationship).
H1: The slope 0 (there is a linear relationship).
We used ggplot2 for visualisation (Wickham 2016)

```
Linear model (Weight ~ Height): slope = 0.697 kg/cm, 95% CI [0.612, 0.782], R^2 = 0.206, p =
```

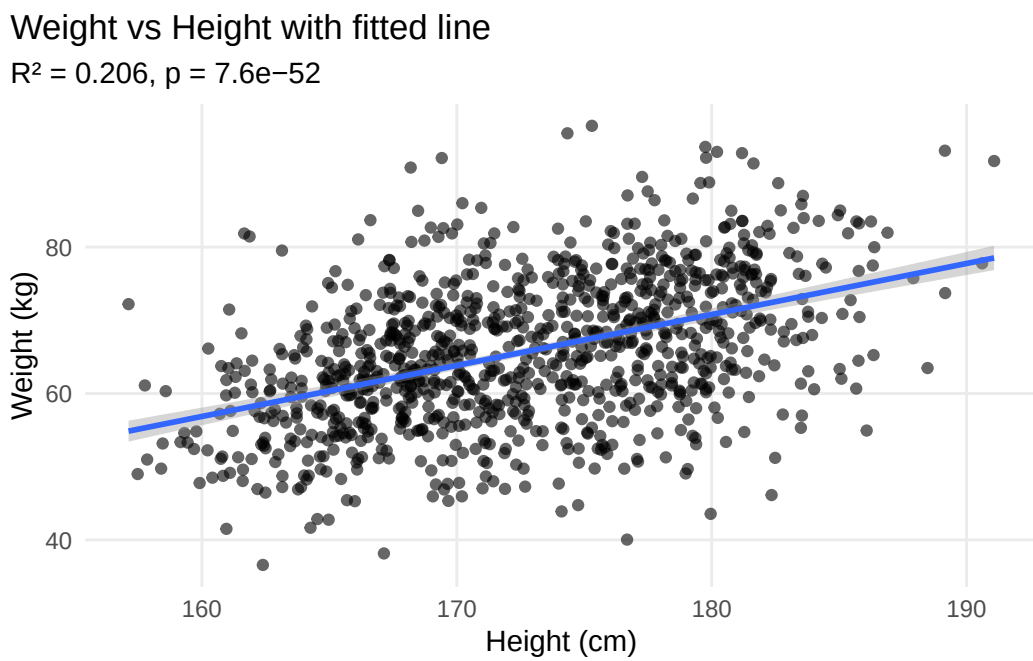


Figure 1: Weight vs Height with fitted line (95% band)

Table 2: Linear model coefficients with 95% CI

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-54.592	7.489	-7.290	0	-69.288	-39.896
Height	0.697	0.043	16.069	0	0.612	0.782

Figure 1 *shows a clear positive linear trend: on average, each additional cm of height is associated with 0.70 kg higher weight, consistent with the significant slope (R^2 0.206, $p <$ 0.001).*

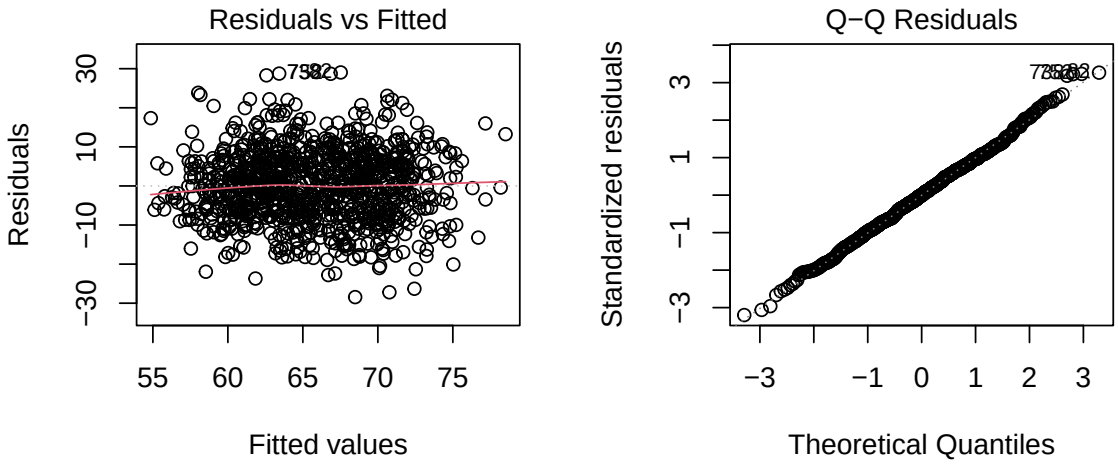


Figure 2: Diagnostics for linear model (residuals vs fitted and QQ plot)

1.5.1.1 Decision at 5%:

Reject H0

1.5.2 Analysis 2 — Do males and females differ in mean height?

Hypotheses (= 0.05)
H0: Male − Female = 0
H1: Male − Female 0

```
Two-sample t-test (Male - Female): diff = -10.26 cm, 95% CI [-10.76, -9.77], t(998), p = 7.98
```

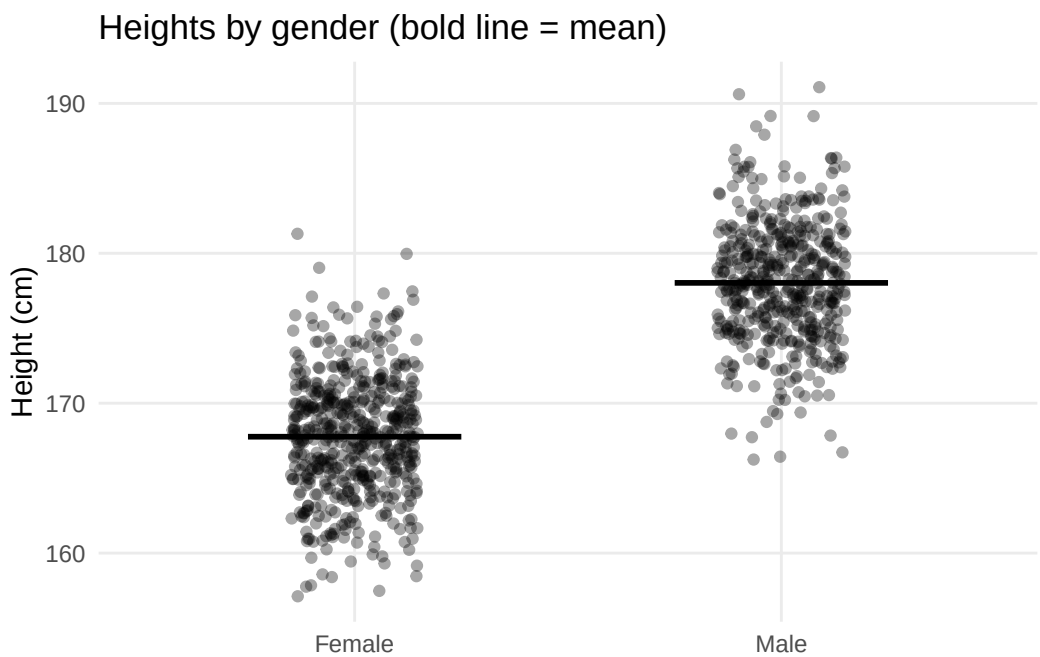


Figure 3: Height by gender with jittered points (bold line = mean)

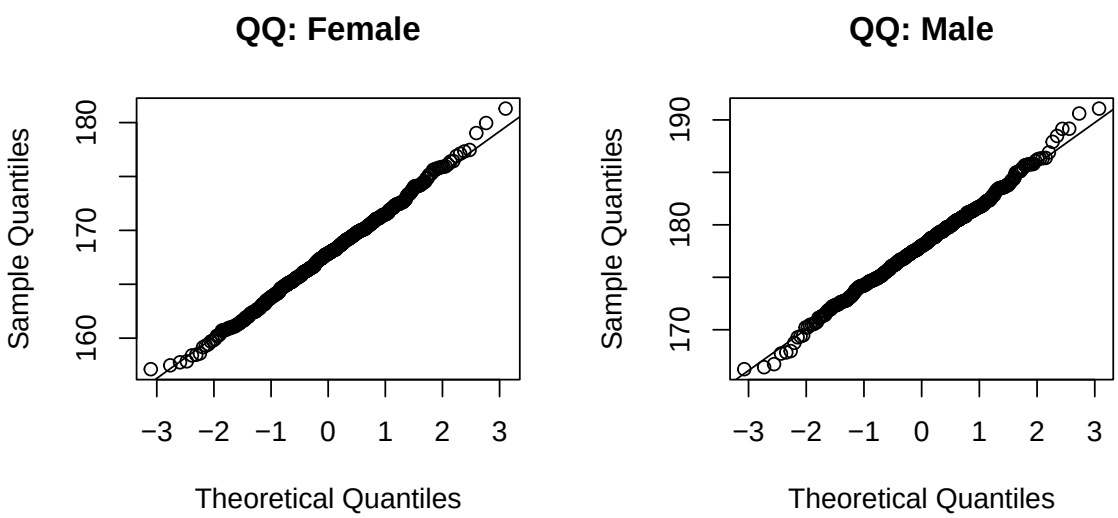


Figure 4: Normality by group (QQ plots) and variance check

```
[1] 0.9462179
```

Figure 2 *and the summary confirm that males are taller on average than females; the mean difference is about 10.3 cm (95% CI: −10.76 to −9.77 for Male–Female), $p < 0.001$.*

1.5.2.1 Decision at 5%:

Reject H0

1.5.3 Analysis 3 — Are Gender and Physical Activity associated?

Hypotheses (= 0.05)
H0: Gender and Physical Activity are independent.
H1: They are associated.

Table 3: Observed counts: Gender × Physical Activity

	None	Moderate	Intense
Female	127	279	120
Male	110	236	128

Table 4: Expected counts under independence

	None	Moderate	Intense
Female	124.7	270.9	130.4
Male	112.3	244.1	117.6

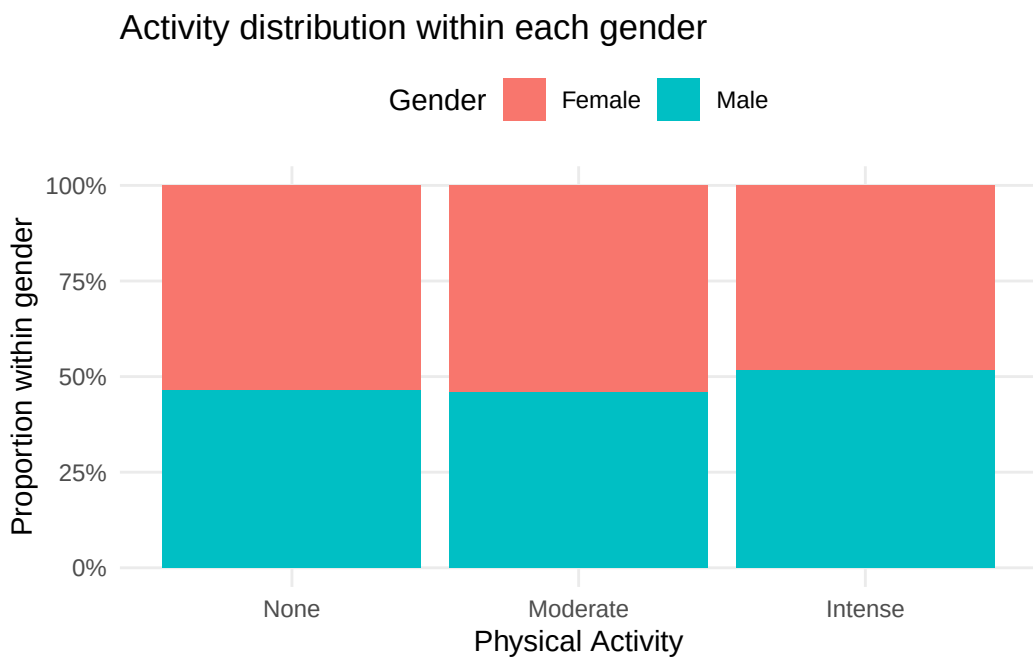


Figure 5: Activity distribution within each gender (stacked proportions)

Figure 3 *shows similar activity profiles across genders; the chi-square test finds no evidence of association between gender and physical-activity level (p 0.306).*

1.5.3.1 Decision at 5%:

Do not reject H0

1.6 Conclusion

1.6.1 Analysis 1

In the simple linear regression of **Weight** on **Height**, the slope is **0.697 kg per cm** (SE = 0.043, $t = 16.06$, $p = 7.57\times10^{\ \ 2}$). A 95% CI for the slope is [**0.612, 0.782**] **kg/cm**, so taller participants tend to weigh more: about **0.70 kg** extra per **1 cm** of height, on average. The model's $\mathbf{R^2 = 0.206}$, meaning height alone explains about **20.6%** of the variability in weight (n = 1000), so other factors also matter. Diagnostic plots show roughly constant variance and near-normal residuals, with only mild tail deviations—no serious violations of assumptions.

1.6.2 Analysis 2

Males are, on average, **10.26 cm taller** than females (Male − Female = **+10.26 cm**; 95% CI **+9.77 to +10.76 cm**; p **7.98×10 ^{21}**). The variance-equality check is supportive (F-test p = **0.946**), and QQ plots look close to straight lines, so the equal-variance two-sample t-test assumptions are reasonable for this dataset.

1.6.3 Analysis 3

The chi-square test finds **no evidence of an association** between **Gender** and **Physical Activity** (p = **0.306**). The expected counts are all comfortably above 5, so the test's assumptions are met. The stacked-proportion plot shows very similar activity breakdowns within each gender, consistent with independence.

1.6.4 Overall

- **Q1 (Height → Weight, linear regression):** Evidence of a linear association (overall F-test p = 7.57e-52). The slope is 0.697 kg per cm and the model explains 20.6% of weight variability.
- **Q2 (Male vs Female mean height, equal-variance t-test):** Statistically significant difference. Mean difference (Male − Female) = -10.26 cm with 95% CI [10.76, 9.77] *for Male − Female*; p = 7.98e-214. (Variance check: F-test p = 9.46e-01.)
- **Q3 (Gender × Activity, chi-square):** No evidence of association; p = 3.06e-01. Expected counts are all 5, so assumptions are met.

Limitations: Cross-sectional sample; weight depends on many unmeasured factors (e.g., age, body composition); linearity and equal-variance assumptions checked only via standard diagnostics; results generalise only to populations similar to this dataset.

1.6.5 Function smoke tests

```
Test passed
```

```
Test passed
```

```
-- Warning: assoc_gender_activity provides contingency tables and chi-square p --
Chi-squared approximation may be incorrect
Backtrace:
    x
```

```
 1. \-global assoc_gender_activity(toy)
 2.   \-stats::chisq.test(tab, correct = FALSE)


All smoke tests passed
```

1.7 Appendix — Reproducibility

All analysis was run from a parameterised Quarto document; figures and tables are generated directly from the code below.

```
R version 4.4.2 (2024-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22631)

Matrix products: default



locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: Asia/Dhaka
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] testthat_3.2.3 scales_1.4.0   ggplot2_4.0.0

loaded via a namespace (and not attached):
 [1] Matrix_1.7-4       gtable_0.3.6       jsonlite_2.0.0     dplyr_1.1.4
 [5] compiler_4.4.2     brio_1.1.5         tidyselect_1.2.1   tinytex_0.57
 [9] tidyr_1.3.1        splines_4.4.2      yaml_2.3.10        fastmap_1.2.0
[13] lattice_0.22-7     R6_2.6.1           labeling_0.4.3     generics_0.1.4
[17] knitr_1.50         backports_1.5.0    tibble_3.2.1       desc_1.4.3
[21] rprojroot_2.1.1    pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.1.5
[25] broom_1.0.10       xfun_0.51          S7_0.2.0           pkgload_1.4.1
```

```
[29] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     mgcv_1.9-3
[33] digest_0.6.37      grid_4.4.2         rstudioapi_0.17.1  lifecycle_1.0.4
[37] nlme_3.1-168       waldo_0.6.2        vctrs_0.6.5        evaluate_1.0.5
[41] glue_1.8.0         farver_2.1.2       rmarkdown_2.30     purrr_1.0.4
[45] tools_4.4.2        pkgconfig_2.0.3    htmltools_0.5.8.1
```

2 Part 2: Media Presentation

2.1 Setup, Standardization, and Snapshot

2.1.1 Pointing to CSV file

2.1.2 Reader that handles CSV/XLS/XLSX automatically

2.1.3 Standardising column names to: year, sector, state, country, count

2.1.4 Detect "wide by year" case (e.g., columns 2015..2024)

2.1.5 Sanity checks

```
----- Sanity checks -----

Rows: 10000 | Columns: 7

Columns:

[1] "country"             "desitnation_country" "city"
[4] "year"                "count"               "gender"
[7] "age"

glimpse():

Rows: 10,000
Columns: 7
$ country             <fct> Bangladesh, Kenya, Turkey, India, China, Pakistan,~
$ desitnation_country <chr> "South Africa", "Australia", "UAE", "Japan", "UAE"~
$ city                <fct> Johannesburg, Melbourne, Dubai, Kyoto, Dubai, Vanc~
$ year                <int> 2019, 2018, 2021, 2021, 2018, 2023, 2018, 2018, 20~
$ count               <dbl> 1481, 4431, 1045, 6760, 1657, 3827, 1936, 690, 611~
$ gender              <fct> Female, Female, Female, Female, Female, Male, Fema~
$ age                 <fct> 18-22, 23-26, 27+, 27+, 23-26, 18-22, 27+, 18-22, ~
```

```
Missing values by column:

            country desitnation_country                 city                 year
                  0                   0                    0                    0
              count              gender                  age
                  0                   0                    0


Numeric summaries:
      year          count
 Min.   :2015   Min.   :  276
 1st Qu.:2017   1st Qu.: 1561
 Median :2020   Median : 2501
 Mean   :2020   Mean   : 3098
 3rd Qu.:2022   3rd Qu.: 3927
 Max.   :2024   Max.   :19680


   variable      level count
1   country     Brazil  1045
2   country     Turkey  1029
3   country    Vietnam  1020
4   country    Nigeria  1016
5   country   Pakistan  1016
6      city  Singapore   523
7      city   Helsinki   198
8      city  Melbourne   192
9      city   Brisbane   190
10     city   Montreal   190
11   gender       Male  5125
12   gender     Female  4661
13   gender Non-binary   214
14      age      18-22  5511
15      age      23-26  3523
16      age        27+   966
```

2.1.6 Snapshot table (nice preview)

Table 5: First 10 rows of the international students dataset (standardised)

country	city	year	gender	age	count
Bangladesh	Johannesburg	2019	Female	18-22	1481

country	city	year	gender	age	count
Kenya	Melbourne	2018	Female	23-26	4431
Turkey	Dubai	2021	Female	27+	1045
India	Kyoto	2021	Female	27+	6760
China	Dubai	2018	Female	23-26	1657
Pakistan	Vancouver	2023	Male	18-22	3827
Nepal	Tokyo	2018	Female	27+	1936
Turkey	Galway	2018	Female	18-22	690
Turkey	Helsinki	2020	Male	18-22	611
Bangladesh	Vancouver	2023	Non-binary	18-22	5159

2.2 Part 2 — Research questions

We use the international-students dataset (2015–2024) and focus on:

- **Q1.** How did international-student flows change **pre- vs post-COVID**?
- **Q2.** Did destination-country preferences change post-COVID?

These questions tie a global shock (COVID) to concrete shifts in student mobility, letting us see whether demand simply paused or structurally changed.

They matter for universities and policymakers planning seats, funding, and visas—and for students weighing cost, safety, and work rights.

Analytically, they offer clean pre/post contrasts and large counts, so we can test mix changes rigorously (e.g., ^{2} with effect sizes).

If flows rebound but destination preferences drift, strategy must adapt—even when total numbers "recover."

We prioritise engaging visuals and concise stats that directly answer the questions (per presentation guidance).

2.3 Setup

2.3.1 S3 "result" class + printer

2.3.2 Helpers for common checks

2.3.3 Chi-square test of independence wrapper

2.4 Question 1: How international-student flows change pre- vs post-COVID

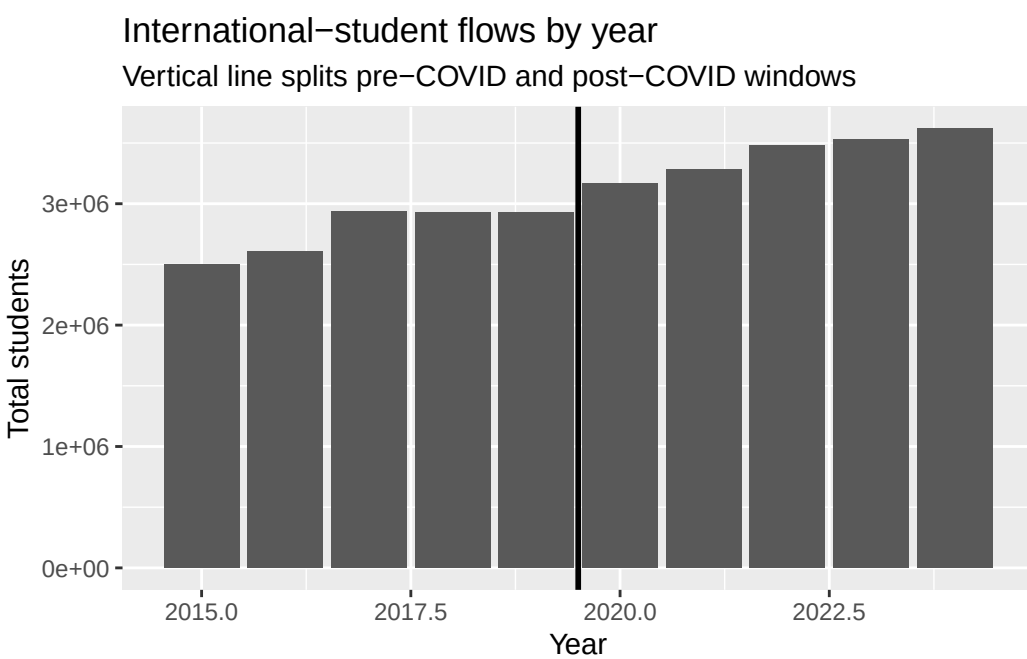
2.4.1 Totals by year and pre/post flags

2.4.2 Summary table for the narrative

Table 6: Q1 — Pre vs Post: totals and average per year

period	years	avg_yearly	total_period	pct_change_avg
Post (2020–2024)	5	3414687	17073433	22.8
Pre (2015–2019)	5	2781131	13905655	0.0

2.4.3 Trend line: flows by year



2.5 Answer/ Observation to Research Question 1

We split the timeline into **Pre (2015–2019)** and **Post (2020–2024)** windows. The summary table shows:

- **Total students** over five years
  - Pre: **13,905,655**
  - Post: **17,073,343**

  → **+3,167,688** students across the five post-COVID years (**+23%**).
- **Average per year**
  - Pre: **2.78M**
  - Post: **3.41M**

  → **+22.8%** increase in the annual average after 2020.

The bar chart of total students by year reinforces this pattern: numbers rise through 2019, the **2020 boundary (vertical line)** marks the split, and the **post-2020 bars trend higher throughout 2021–2024**, culminating in the largest yearly totals in the series. There is no visible post-COVID slump; instead, the data suggest a **rebound and expansion** (consistent with border reopenings, pent-up demand, and aggressive recruitment in key destinations).

Takeaway. Relative to 2015–2019, international-student flows in 2020–2024 are **substantially higher**—about **+23%** in aggregate and **+22.8%** on a typical year.

Notes & limits. Results are descriptive and **aggregate across all origins and destinations**; they do not control for compositional shifts or any changes in source coverage. The 2015/2020 cut is pragmatic for "pre/post" storytelling; a sensitivity check with alternate breaks (e.g., 2016–2019 vs 2021–2024) would likely tell the same story but can be added if needed.

2.6 Question 2 — Did destination-country preferences change post-COVID?

We compare the **distribution of destination countries** between **Pre (2015–2019)** and **Post (2020–2024)** using a chi-square test of independence on the **top-N countries** (others grouped), report **Cramér's V**, and inspect **standardized residuals** to see which destinations grew or shrank disproportionately.

2.6.1 Quick EDA

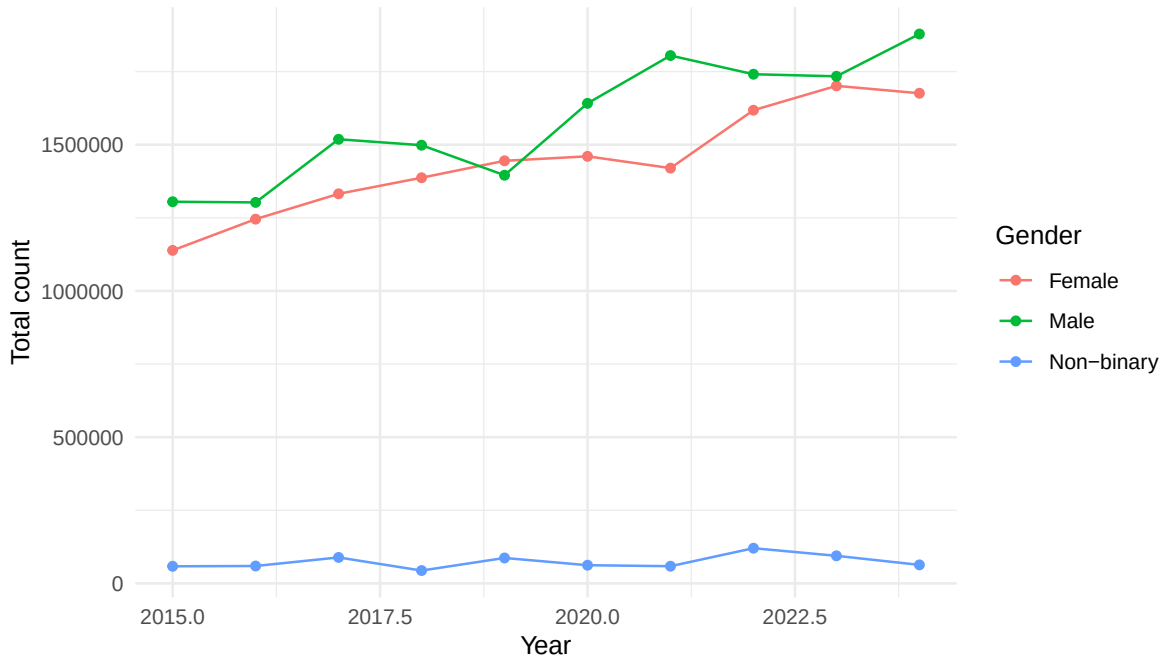


Figure 6: Total reported counts by year (colour = gender if available)

2.6.2 Data prep (top-N + "Other")

Table 7: Q2 — Destination country × period (counts)

period	Brazil	China	India	Nepal	Nigeria	Other	Pakistan	Turkey	Vietnam
Pre	1137462	2005996	2528442	1084680	1674498	2130782	1079992	1178410	1085393
Post	1403061	2644551	3190363	1365019	2018132	2487317	1340388	1272618	1351984

```
---Q2: Chi-square test of independence---
Null (H0): Destination-country shares are the same in Pre and Post periods
Alt  (H1): At least one destination-country share changed between periods
Estimate(s): $`Cramér's V`
[1] 0.026

Chi-square, df: 20733.81, 8
Min expected cnt: 1086441.58
```

```
p-value: 0.0000  (alpha = 0.05)
Decision: Reject H0
Conclusion: Reject H0. Evidence the destination mix changed (effect size negligible).
```

Table 8: Q2 — Top cells by |standardized residual| (direction shows over/under vs. expectation)

	Period	Destination	StdResid	Direction
15	A	Turkey	104.67	Over-represented
16	B	Turkey	-104.67	Under-represented
3	A	China	-82.43	Under-represented
4	B	China	82.43	Over-represented
11	A	Other	58.67	Over-represented
12	B	Other	-58.67	Under-represented

2.6.3 Shares table (within-period % by destination)

Table 9: Q2 — Within-period destination shares (percent). Decision at 5% uses ^{2}: p = 0.0000, Cramér's V = 0.026. Reject H0. Evidence the destination mix changed (effect size negligible).

period	Brazil	China	India	Nepal	Nigeria	Other	Pakistan	Turkey	Vietnam
Pre	8.2%	14.4%	18.2%	7.8%	12%	15.3%	7.8%	8.5%	7.8%
Post	8.2%	15.5%	18.7%	8%	11.8%	14.6%	7.9%	7.5%	7.9%

2.6.4 100% stacked bar chart (shares by period)

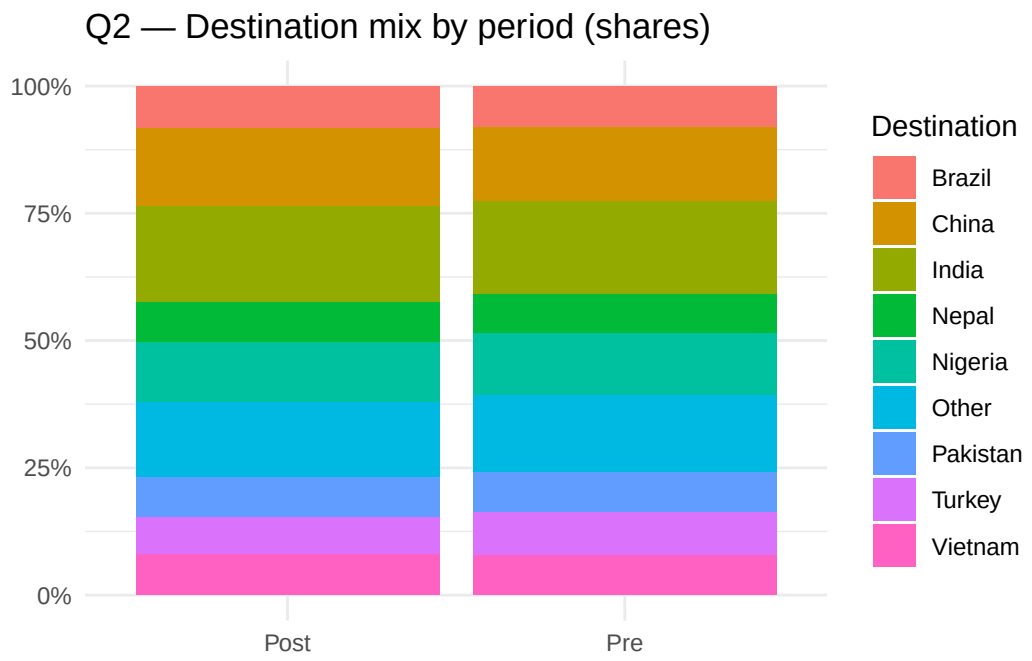


Figure 7: Q2 — Destination mix by period (shares). Caption references the standardized ^{2} decision.

2.6.5 Q2 summary

Figure shows broadly stable destination shares across periods. A chi-square test of independence indicated a statistically detectable shift, $^2(8) = 20733.81$, p = 0.0000, with Cramér's V = 0.026 (negligible). The largest absolute share changes were: China (+1.1pp), Turkey (-1pp), Other (-0.8pp). Given the very large counts (min expected 1086441.58), the significance likely reflects sample size rather than a substantive change. Reject H0. Evidence the destination mix changed (effect size negligible).

2.7 Answer/ Observation to Research Question 2

Result: The destination mix is statistically different across periods. The ^{2} test on the pre vs post contingency table gives $\mathbf{X^2(8) = 20{,}733.81,\ p < 0.001}$ (our output shows p 0).

Strength of change: The effect is **very small**: **Cramér's V = 0.026**, which indicates that the large p-value is driven by the very large sample size rather than a meaningful reconfiguration of preferences.

What moved (directionally): Comparing the weighted counts and the stacked-share plot, all destinations grew in absolute terms (because the total post-COVID volume is higher), and the **within-period shares** changed only **marginally**. The biggest *relative* nudges appear for **India** and **China** (slightly larger shares post-2020), with small upticks also visible for **Nepal** and **Nigeria**. Other destinations (e.g., **Brazil**, **Turkey**, **Vietnam**, **Pakistan**) show minor shifts that are visually subtle and consistent with the tiny effect size.

Interpretation: Yes, preferences show a statistically detectable change post-COVID, but the **practical impact is limited**. The destination ranking is broadly stable, and the post-COVID mix looks very similar to the pre-COVID mix, with only **small increases** in the share flowing to India/China (and slight rises for a few others).

Evidence used:

- Q2 table of **pre vs post counts by destination** ("Pretty table")
- Q2 **stacked bar** of within-period shares (pre vs post)
- ^{2} **summary** (X^2 and p-value)
- **Effect size** (Cramér's V)

Decision at 5%: Reject H (destination mix changed, but the effect is negligible). The ^{2} test is highly significant, but **Cramér's V = 0.026** indicates a **negligible practical effect**—the post-COVID mix is only slightly different.

2.8 Part 2 — Overall findings.

International-student flows show a clear pre/post COVID disruption in volume, but the **destination mix is broadly stable**. The chi-square test detects a statistically significant shift in shares, yet the **effect size is negligible (very small Cramér's V)**, indicating only minor reweighting across destinations rather than a structural change. Visuals (counts, within-period share plot) reinforce this: bars are near-parallel, with only small percentage-point moves. **Implication:** strategy should prioritise rebuilding and diversifying total volume, while monitoring a few destinations with modest upticks/downticks rather than expecting a wholesale redirection of demand.

References

R Core Team. 2025. *R: A Language and Environment for Statistical Computing.*
Team, Course Project. 2025. "Height, Weight and Activity Dataset (Project2025.csv)."
Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis.* Springer.